

Transformer models have become the default architecture for large-scale language processing. Residual connections and layer normalization stabilize training at depth. Paragraph 1.1 continues with a second sentence about attention.

Attention computes a weighted combination of values. Scaled dot-product attention remains the most common building block in encoder and decoder stacks. Paragraph 1.2 continues with a second sentence about attention.

Positional encodings inject order information that self-attention cannot recover from the input tokens alone. Relative variants later improved length generalization. Paragraph 1.3 continues with a second sentence about attention.

Empirical results on translation and language modeling established the recipe used by later large language models. The same block appears in vision and speech systems. Paragraph 1.4 continues with a second sentence about attention.

Transformer models have become the default architecture for large-scale language processing. Residual connections and layer normalization stabilize training at depth. Paragraph 2.1 continues with a second sentence about attention.

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Transformer models have become the default architecture for large-scale language processing. Residual connections and layer normalization stabilize training at depth. Paragraph 3.1 continues with a second sentence about attention.

Attention computes a weighted combination of values. Scaled dot-product attention remains the most common building block in encoder and decoder stacks. Paragraph 3.2 continues with a second sentence about attention.

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